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The Complete Surface and Subsurface Redevelopment of a Large Field in South of Sultanate of Oman

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Abstract

The field in Hand, which is one of the largest fields in south of Sultanate of Oman, has been producing since the 1980s. The field has gone through several development phases starting with depletion (both horizontal and vertical developments), steam trials, waterflood and ending up with a polymer flood rotation.

After the water flood FDP, polymer phase 1 implemented the injection of HPAM 3630S polymer solution with a viscosity of 15 cP into 26 water injectors in the main reservoir and 1 in a secondary reservoir as a pilot. Followed by phase 2 polymer consisting of an extra 19 injectors covering more geographical areas of the field (Svec et al,2021)

A complete redevelopment of the field was sanctioned in 2017, where a new field development plan called for intensive infill drilling that will result in doubling the field's well count to reduce the water flood well spacing from 600 m inverted 9-spot to 424 m inverted 9-spot in the flank and converting inverted 5-spot to inverted 9-spot in the valley at 424 m spacing, with no expansion of the existing polymer Plant capacity and flooding through what is called polymer rotation that is explained in the paper. This project is expected to add reserves of about 8% to the total recovery of the field which is a huge addition given the size of the field making it very attractive economically.

Additionally, there is a significant scope of converting the corner producers (in existing patterns) to injectors and rotating polymer injectors to other patterns. These conversions are linked to the surface readiness (i.e. flowlines and overhead lines need to be in place) and the subsurface (enough offtake wells are in place). A delay in the conversion plans will jeopardize future drilling as the old infrastructure is often clashing with new well locations.

Lastly, there is a considerable number of active horizontal wells which pose some risks for the future development through interference, conformance, or collision risk during drilling. Some of these horizontal wells will have to be abandoned before the new development wells can be drilled.

To make the subsurface re-development possible, the project requires a complete new urban plan where new infrastructure for the off-plot such as flowlines, overhead lines, and roads, and on-plot liquid handling, electrical infrastructure and pumps to be installed. For the newly built infrastructure, the old clashing infrastructure will be demolished. Existing wells will be changed over to the new infrastructure in steps to

ensure deferment is minimized. The demolition of the old infrastructure will also ensure the rig can access all the new well locations.

All these interfaces are being managed by an integrated team that covers the surface off-plot, on-plot, subsurface and drilling. The main project management challenge is that all the activities are linked and need to be delivered in timely fashion to make sure the drilling activities can be executed without delay. Key focus areas are executing the work safely, managing SIMOPS activities, minimizing the deferment and making sure all activities are delivered in a timely fashion to ensure the full value for the project can be realized.

Introduction

The field which was Discovered in 1958 on the southeastern flank of the Sultanate of Oman Basin, was initially deemed commercially non-viable based on the poor performance of five exploration wells. In 1969, Petroleum Development Sultanate of Oman (PDO) acquired the concession license, and further drilling activities commenced in 1976. Commercial oil production from the field began in 1980. Figure 1 illustrates the geographical location of the field within Sultanate of Oman.

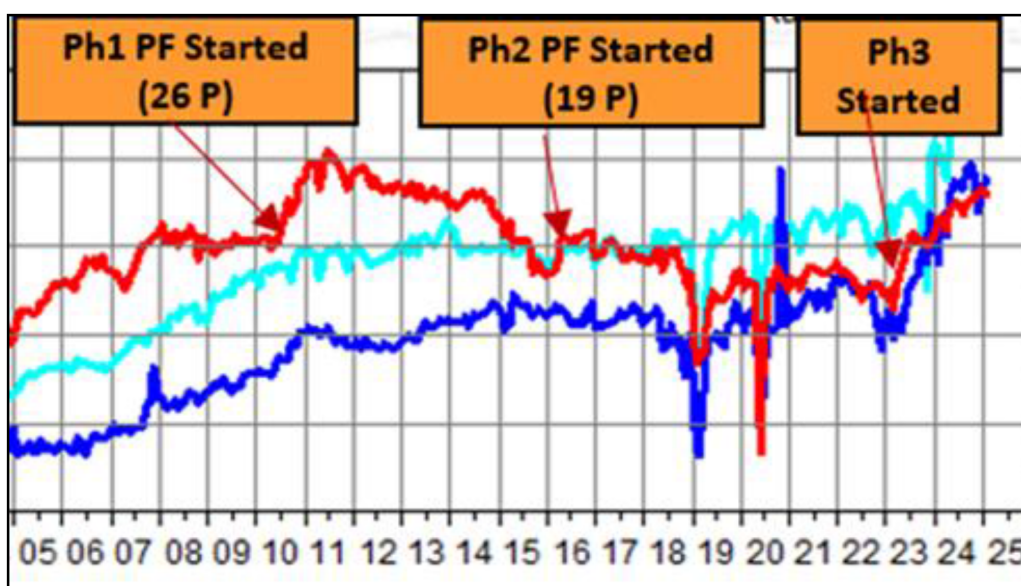


Figure 1—Historical Production of the field throughout different development phases (oil-red, Liquid-cyan, Injection-Blue)

The target reservoir is characterized as a multi-Darcy permeability system connected to a weak underlying aquifer. The in-situ oil is moderately heavy, with an API gravity of approximately 22° and a viscosity of 90 cP. Initial reservoir pressure was around 9300 kPa at a depth of 610 meters TVDSS, with a bubble point pressure estimated at 7000 kPa.(Al Sawafi et al. 2023)

Enhanced oil recovery (EOR) efforts began as early as 1985, with pilot tests for both steam and polymer flooding. Waterflooding was initiated in 1991 using vertical injection wells and expanded in 1996 with the addition of more vertical injectors. Horizontal injection wells were introduced in 1999 to further improve sweep efficiency. Based on encouraging results from the polymer flood pilot and subsequent evaluations, a full-field development plan (FDP) for polymer injection was approved in 2006. Commercial-scale polymer injection, involving 23 vertical and 3 horizontal wells, commenced in 2010 with a target polymer viscosity of approximately 20 cP. A second phase of polymer flooding began in 2015, and the third phase is currently in progress. Alkaline-Surfactant-Polymer (ASP) injection has also been identified as a future strategy to enhance oil recovery.

The new infill development -called Phase 3- includes an intensive infill drilling campaign aimed at reducing waterflood well spacing from 600 meters to 424 meters (inverted 9-spot configuration) in flank

areas, and converting inverted 5-spot patterns to inverted 9-spot in valley regions at the same spacing. Additionally, Phase 3 plans to utilize the existing Polymer Plant for polymer rotation across the field, without further expansion of the existing plant's capacity.

The figure below shows the fields historical production at different development stages with clear emphasis on the benefit if polymer floods in arresting the decline without additional wells. It is also clear the increase due to the start of the infill drilling in 2023.

Subsurface Re-development

Infill Drilling

The development mainly revolves around an extensive infill drilling campaign comprising over 300 oil producers to be executed in a span of 4 years, the majority of which are equipped with Beam Pumps (BP) to optimize artificial lift performance. A subset of producers is completed with Electrical Submersible Pumps (ESP) integrated with Downhole Gauges (DHG) to enable real-time pressure surveillance. Additionally, select ESP wells are equipped with Y-tools to facilitate wireline or memory logging operations without the need for pump retrieval, which are in corners of patterns to ensure full field coverage. Complementing the patterns, a series of water injectors has been drilled and completed with a "fat" well design to accommodate enhanced surveillance and future re-entry. These injectors are instrumented with distributed fiber optic cables to enable continuous conformance monitoring through temperature profiling and potential distributed acoustic sensing (DAS) applications, supporting improved sweep efficiency and reservoir management (al Hashmi, 2022).

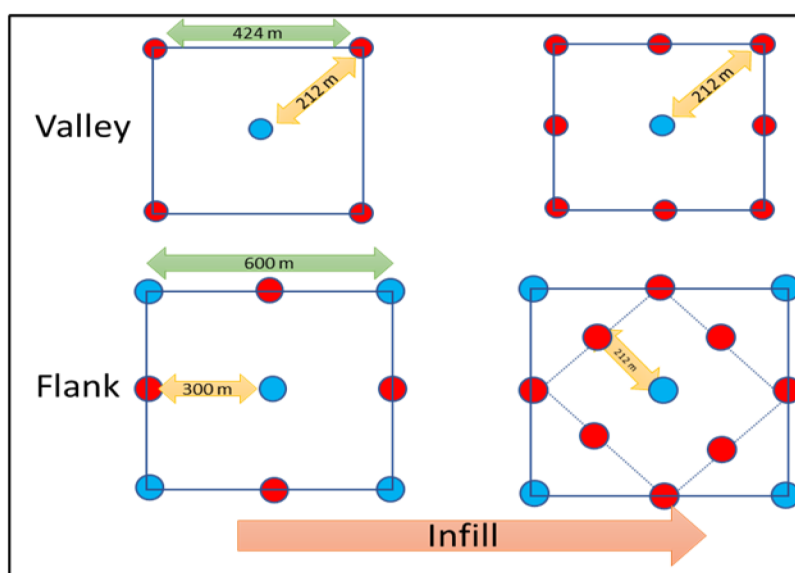


Figure 2—The new pattern configuration

The drilling was being carried out by a well engineering integrated project management model, in which the wells are delivered by the contractor with a lump sum. The drilling activities reached to a pace where there were 4 rigs operating at the same time in an area of 1 km², which was made feasible via the new road and street design of the field. The drilling was divided in 4 different blocks determined by surface layout, where every block had its own network of roads, overhead lines and MSVs that enabled the wells hookup and production as soon as the drilling was complete.

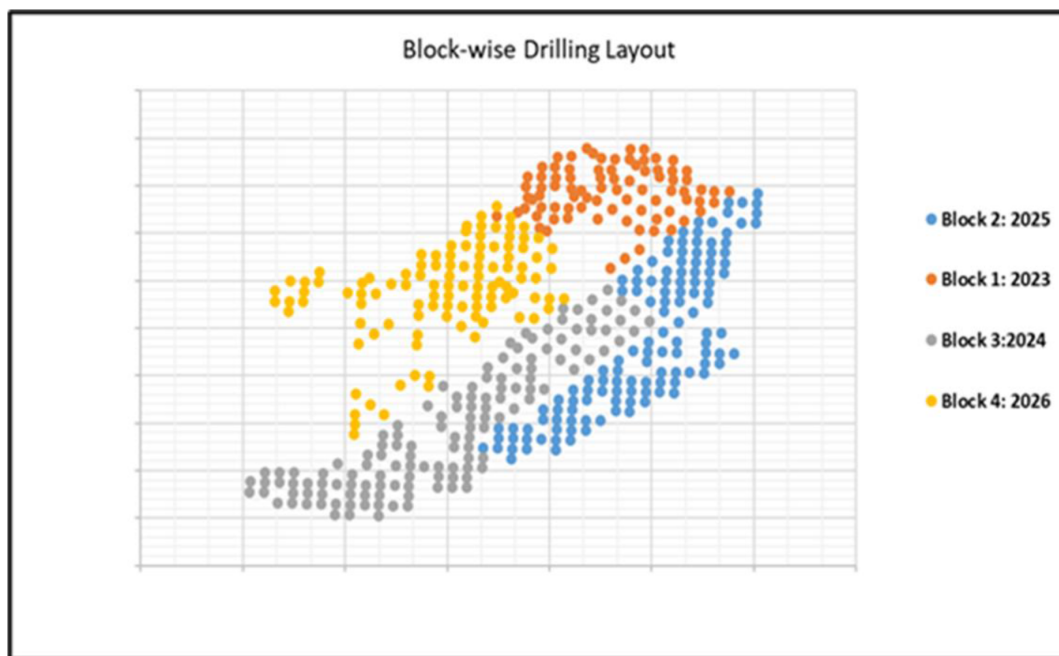


Figure 3—Division of the wells per surface blocks per year

Conversion

Phase 3 Infill drilling reduces the well spacing from 600m to 424m creating more patterns in the field which require more injection. As well as infill drilling, it calls for conversion of every corner producer (in the existing 600m spacing pattern) to an injector to create a uniform inverted 9-spot pattern. The spacing between the injector and the producer reduces to 212 m as shown in the figure. The main purpose of conversion is mainly to support and enhance the pressure in the new patterns. The main priority of the conversion is where the infill drilling started. However, few conversions are optimized and delayed since they are producing high net oil. In these patterns, the pressure is being monitored in order to avoid any sudden drop in reservoir pressure. Based on the FDP, 53 wells are planned for the conversion to injectors aligned with the drilling sequence. For patterns where conversion is required, well conversion will occur as soon as the infill drilling is completed. Locations of the conversion candidates are prepared for the conversion activities under off-plot construction scope. Subsurface scope of the conversion will be handled by dedicated hoist to ensure the alignment with drilling activities. After converting oil producers to injectors, oil production will decrease immediately. However, the oil will be recovered from the new infill wells after stable injection of few months.

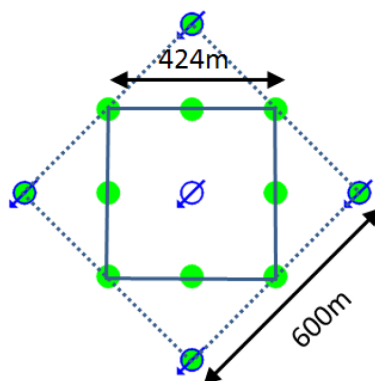


Figure 4—Illustration of corner wells to be converted to injectors after drilling

Abandonment

Historically, around 140 horizontal wells were drilled in the field and most of them are still active. The existence of the horizontal wells in a field where vertical wells are planned could pose some risks like interference, conformance or collision risk during drilling. Consequently, two scenarios have been considered to deal with the existing horizontals as the infill project is executed:

1. Keep the horizontal producers active
2. Abandon the horizontal producers when the vertical infill wells are drilled.

The key finding of abandoning current horizontal producers leads to a decline in production and a reduction in net present value (NPV). Additionally, it causes a loss in reserves recovery because the horizontal wells access valuable oil zones not fully reached by the new vertical infill wells. Nevertheless, it is important to emphasize that the infill development remains profitable. Therefore, it was recommended to maintain as many of the existing horizontal producers as possible (scenario 1), with individual evaluations conducted on a case-by-case basis through pattern reviews and ad hoc well assessments, at least annually. It has been agreed to retain horizontal wells producing high net oil, while monitoring their BSW to determine the appropriate abandonment timing. When considering the drilling of injectors or converting producers to injectors, the proximity to existing horizontal producers must be carefully evaluated to avoid potential water short-circuiting through the horizontal wells, which could result in production and reserves loss. (Al Ghailani et al, 2024)

Polymer Rotation

It was determined that the development would be through actively rotating polymer injection across the patterns. This rotation utilizes the existing polymer plant capacity of 27 t/d instead of expanding the plant. This concept stops polymer injection into the first two phases Phase 1 and 2 patterns -highlighted in the below map- prematurely to redirect injection towards higher incremental recovery and value from new patterns. Once polymer injection is transferred to a new pattern, the existing pattern is converted back to a water flood.

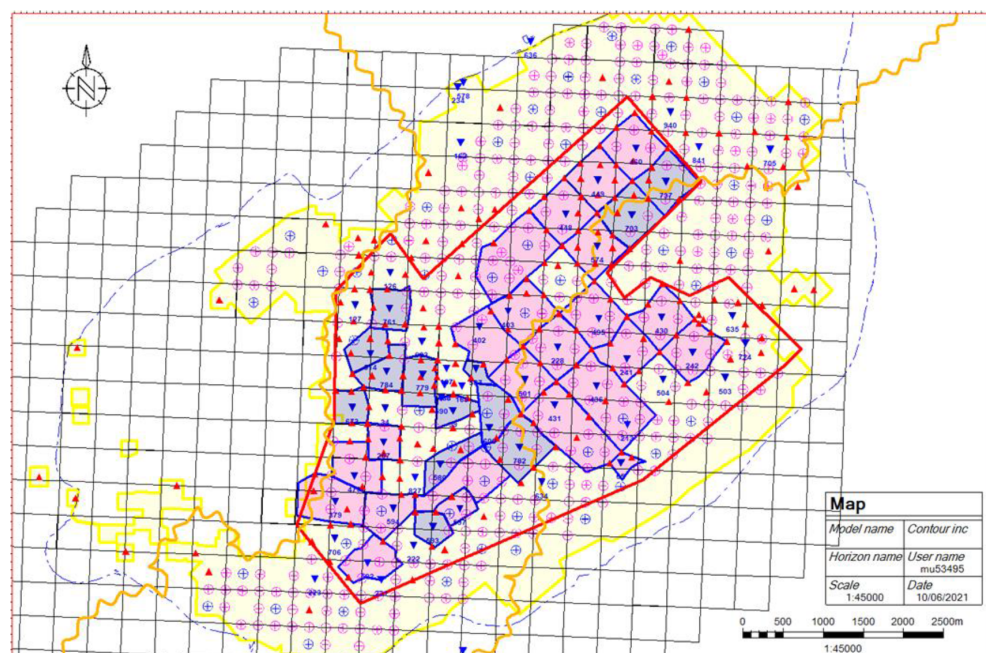


Figure 5—The Field Polymer Patterns [Red Polygon] prior to the infill drilling project

The polymer rotation scheme recommended that learnings from Phase 1 and 2 should be used for further WRM optimization of the rotation strategy. The new phase 3 polymer patterns will be kept on injection for a minimum of 2 years, based on learnings from phase 1 & 2 patterns. The polymer pattern response time to polymer flood were observed to vary from a few months to 1.5 years. Thus 2 years of polymer injection is required at minimum to allow to assess polymer response confidently. Once the 2-year window pass then an assessment of the pattern would be done through a pre-set PUF threshold in which if the patterns passed it would continue injecting, otherwise it would be turned to a waterflood pattern, and the polymer capacity would be moved to a new candidate pattern.

Surface Redevelopment

Facility Expansion

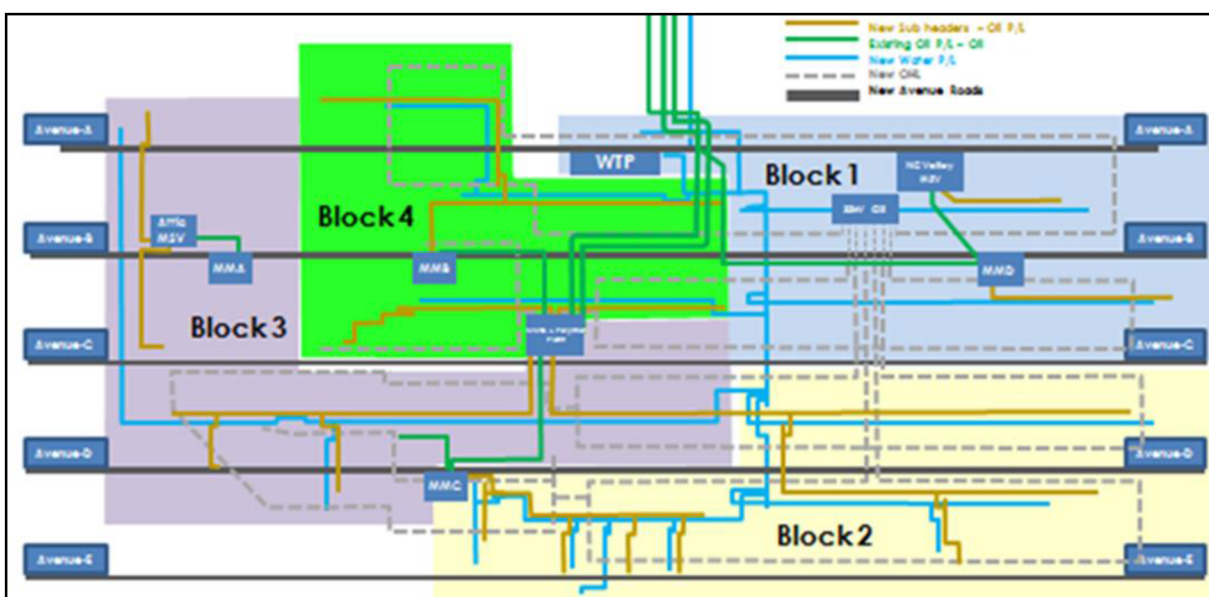


Figure 6—Dividing the field into surface blocks

New Urban Plan

The Project area (the field span of 6 km × 10 km) is very congested with existing wells, flowlines, Pipeline and OHLs. It is not possible to drill and hook up around 500 incremental of producer wells and other associated water injection/DWD wells without a major redevelopment of the surface infrastructure. Therefore, the urban plan has been developed, which consists of a fish bone structure of headers and sub-headers in the field which will be fed to different Gathering stations and the main production station respectively. The Urban Plan mainly consists of row of wells, main header corridor, sub header corridor, flowline corridor, OHL corridor and rig access road. Sub header corridor is planned between two avenue road and 4 rows of wells (existing and new) located in-between avenue road and sub headers, i.e. eight rows of wells in-between two avenue road and sub header in the middle. For collection of fluid from production wells, Multisport selector valves (MSV) stations are located on either side of Sub-header corridor which will be connected through bulk headers. In addition to the production there are injection wells which are connected directly to sub header. Urban plan is developed providing space for level 2 wells (future in fill wells). These wells are not part of this project. Only space is allocated for future MSV station, flowlines, sub header and main header.

To minimize the initial cost of MSV stations, wherever practicable the existing well flowlines are segregated and connected to one MSV in MSV station, so that only one MSV required to be installed in

MSV stations in the initial phase of the project. Each sub header corridor and main header has a number of productions, water injection, DWD headers and flow lines as appropriate.

In general, 8 rows of wells located in between two existing avenue roads and parallel to it. Each sub header is installed in the middle of two adjacent avenue roads. Therefore, four nos. of wells will be located on either side of sub header. Ideally four such rows of wells with seven nos. of well counts in each row are termed as MSV module.

In between sub header and avenue road there are generally four rows of wells, in between two rows of wells there is rig road planned, wells on either side rig road can be approached from this rig road. Hence total two rig roads between sub header and avenue road, and four rig road between two avenue roads serving eight rows of wells. To minimize the space requirement for drilling the concept of Master waste pit and Master water pit philosophy is followed. Master waste and water pit strategically located in each of MSV module. Level 2 well locations are allotted for these pits. The urban plan has four types of services i.e. production, water injection, DWD and polymer injection. There are main headers, sub headers, bulk headers and flowline corridors for all these services.

Construction & Changeover

The project comprises several phases focused on surface redevelopment. One of the major phases involves revamping the off-plot urban plan, which aims to transition existing wells into a new, organized infrastructure layout. This includes re-routing current flowlines into a structured pipeline network via MSV stations and gathering stations, ultimately enabling rig access for future infill drilling activities.

Flowline and Pipeline Network.

- Reconfiguration of existing flowlines into a streamlined system to facilitate efficient fluid transport and rig accessibility.
- Installation of carbon steel pipelines with polyethylene lining for production wells, involving the redirection of flowlines from multiple fields into centralized stations.
- Deployment of similar pipelines for water injection wells, combining injectors from different fields into shared headers.
- Establishment of polymer injection lines extending from the main processing plant to designated injector wells.

Facility and Station Upgrades.

- Construction of new MSV stations along with integrated metering facilities to monitor and manage flow.
- Modifications to existing gathering stations to align with the updated flowline network and capacity requirements.

Electrical Infrastructure.

- Implementation of an overhead electrical line system and the setup of a gas-insulated switchgear building.
- Expansion of the switchyard area to accommodate high-voltage transformers and associated power distribution needs.

Support Systems. Installation of essential supporting infrastructure, including electrical systems, telecommunications networks, and control and automation facilities to ensure seamless integration and operation.

Demolition and Decommissioning

Removal of outdated infrastructure, which includes existing flowlines, overhead electrical lines, demulsifier skids, and various associated components that are no longer required under the new layout.

The other phase primarily focuses on the development of on-plot facilities. This includes the installation of a bulk separator and an Enhanced Gas Flotation Tank (EGFT) package, along with a storage tank and several GLR pumps and vessels. Wet crude will be managed through dedicated transfer pumps, while water handling will be enhanced with the addition of new water booster pumps.

Conclusion

The project described is one of the company's largest projects in recent times with extensive capital investments and even bigger production promises. The additional recovery of the project adds an incremental RF of ~8% throughout the 25 years of expected project life. The development today, stands at a 100% completion rate of the surface scope where all the new facilities are up and running and a great pace at drilling with over 70% wells drilled and completed beating the original schedule by over 6 months. Conversions and abandonment are at a rate of nearly 50% where they are being actively optimized through WRFM daily work. Production promises are being well met making it a great success story after years of complexity,

The execution of the project requires a great deal of collaboration between countless number of companies, departments and individuals and the execution team succeeded in managing the interfaces in delivering the milestones safely, timely and cost effectively. The paper aims to demonstrate that even at a very late stage of a brown field's life lies great opportunities to increase recovery and gain attractive and positive cashflow. The project also shows how integration amongst all stake holders and disciplines can be levered to deliver the total scope in a timely and safely manner.

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